

SUBHIKSHA STEPHAN

737-228-5422 · sks111@duke.edu · avidnerd.vercel.app

EDUCATION

Duke University

Biomedical Engineering & Electrical and Computer Engineering, Certificate in Innovation & Entrepreneurship

Expected May 2030

Durham, NC

EXPERIENCE

Inoki Labs

Co-Founder, CTO

May 2026 – Present

Remote

- built an LLM-guided adversarial search engine that discovers rare failure modes in learned robot locomotion policies and automatically retrains against them, outperforming open-source SOTA policies in reducing failure rates
- scaled the pipeline across a 16-dimensional perturbation space and five robot platforms (Unitree Go1/G1, Boston Dynamics Spot, Google Barkour, Booster T1) in MuJoCo MJX, at roughly one day of unattended GPU time per robot
- selected for Antler's VC program, unlocking \$2.1M in compute credits and partner benefits across 150+ companies
- first-authored "Discrimination and Calibration Decouple in Runtime Failure Monitors Under Distribution Shift," accepted to **IEEE MIT URTC 2026**, pending publication in IEEE Xplore

New Jersey Science Research Symposium

Co-Founder, Administrative Manager

Nov. 2025 – Present

- helped secure 81 participants and 21 schools
- organized fair rules and logistics
- developed public website and registration portal for judges, students, and advisors (njsrs.org)
 - **Stack:** Next.js, TypeScript, Tailwind CSS, Firebase, and SendGrid (Vercel)
- assisted in social media marketing for fair promotion and judging outreach (used Canva, Capcut, Adobe Suite)
- in charge of managing fair day tech logistics and tech support throughout the registration process

STEM Outreach Club, Computer Science Club

President, Founder

Feb. 2024 – Present

- organized a community-wide tech drive that refurbished and donated 150+ computers and devices to local families
- founded Catalyst, the school's first computer science magazine; led editing and design on the inaugural edition and served as chief editor for subsequent issue
- Built and delivered 5 websites and apps for local businesses, nonprofits, and student organizations

Essex County Office of Public Health

Volunteer Software Lead

Jun. 2025 – Nov. 2025

- Built a retrieval-augmented chatbot that answers resident questions about county healthcare services, using ChromaDB and Sentence Transformers over 500 pages of county program documentation with LangGraph orchestrating retrieval and response
- Led ~10 student volunteers through the build, splitting the work across ingestion, retrieval, and frontend, and reviewing contributions into a working system
- Deployed to the county's WordPress site via a React widget on a Python/FastAPI backend
- Stack: Python, FastAPI, LangGraph, ChromaDB, Sentence Transformers, Groq, React, WordPress

RESEARCH & PROJECTS

Kessler Foundation - Developed artificial intelligence tool under Dr. Selvan to detect fatigue in cancer patients via EEG signals; AI tool is now actively used to evaluate efficacy of rehabilitation therapy in cancer patients

Massachusetts Institute of Technology - Mentored by Dr. Aarti Krishnan at the Collins Lab to develop an LLM guided methodology for designing de novo antibiotics.

Mount Sinai Hospital - Worked with Dr. Charles Mobbs on a research project that uses artificial intelligence to design drug candidates protective against Alzheimer's Disease and age-related diseases.

AeroFlux (Independent) - Developed a smart inhaler spacer to predict the percentage of delivered inhaler dose and diagnose technique errors to improve quality of life for asthma patients

Compass - Won top 5 at the Corgi Hackathon at YC Startup School

HONORS

- **Terra North Jersey Science Fair (research):** Bioinformatics and Computational Biology - 2nd Place, Association for Computing and Machinery Award, IEEE Young Engineer Award
- **Blue Ocean Student Entrepreneurship Competition Top 10 Finalist (out of ~13,000)**
- **Genes in Space Semifinalist (Top 37 teams out of ~1000)**
- **Penn EcoVenture Challenge 2nd Place-** Winner for a climate tech startup competition hosted by UPenn. Our project was a blockchain based green certification platform with a consumer extension to detect greenwashing.

TECHNICAL SKILLS

Python, Java, Javascript, React, Next.js, Web Development, MuJoCo MJX, PyTorch, Tensorflow & ML Libraries, Git